OA-CXR: An Anatomical-Retention-Guided System for Selective Chest Radiograph Report Review

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ABSTRACT

Vision–language models can draft chest radiograph reports, but a generated statement is trustworthy only if the image presented to the model contains the anatomy needed to support it. Negative statements are especially vulnerable: when preprocessing or cropping removes the costophrenic angles, "no pleural effusion" loses its visual evidence, yet generation confidence is unaffected. We present OA-CXR, an intelligent review-support system that verifies this anatomical evidence statement by statement and allocates clinician review accordingly. OA-CXR combines a knowledge base that maps findings to their diagnostic lung regions with a learned estimator of source-relative anatomical retention, defined as the fraction of a region in the acquired radiograph that remains in the presented image. A two-stage global–regional network, trained with mask-derived targets and auxiliary lung segmentation, estimates retention from the presented image and a region query alone; its estimates drive interpretable statement flags and budget-constrained report prioritization. Across 343,395 queries from four public datasets, OA-CXR achieved the lowest error among nine retention regressors (macro MAE 0.0296). For MAIRA-2 reports with eligible statements, reviewing 50% captured 91.7% of reports containing low-retention statements, compared with 50.0% for token log probability, which performed at chance level; at budgets of up to 30%, OA-CXR matched the oracle. On an independent pediatric cohort of 797 radiographs with clinician-annotated references, the frozen system achieved MAE 0.0281 and AUROC 0.974 without adaptation. Anatomical evidence sufficiency thus provides an image-grounded review signal that confidence-based triage cannot supply.

1. Introduction

Vision–language models (VLMs) now draft complete chest radiograph reports (Bannur et al., 2024; Z. Chen et al., 2024; Sellergren et al., 2025; Zambrano Chaves et al., 2025), and expert evaluations judge many generated reports equivalent or preferable to clinician-written ones, although clinically significant errors persist (Tanno et al., 2025). Deployment will therefore remain supervised, and clinician attention becomes the scarce resource: an intelligent reporting system must decide not only what to write but also which statements need review. Selective prediction and learning to defer address this question (Hendrickx et al., 2024; Mozannar & Sontag, 2020), and deferring selected cases to clinicians has improved combined human–AI accuracy in medical imaging (Dvijotham et al., 2023; Leibig et al., 2022). Such policies, however, are only as effective as the signal that decides what to defer.

A generated statement is trustworthy only if the image the generator received contains the evidence that the statement requires, and this image often differs from the acquired radiograph. Many vision–language pipelines resize the shorter image side and center-crop a square input (Radford et al., 2021), which discards the superior and inferior margins of a portrait radiograph; region-of-interest cropping, image export, and patient positioning can likewise truncate the lung apices, lateral chest walls, or costophrenic angles. Radiographic quality standards require these regions to be visible because they cannot otherwise be assessed (European Commission, 1996). Negative statements are particularly exposed: when the lung bases are missing, "no pleural effusion" becomes an unsupported assertion, and a radiologist would instead report that an effusion cannot be excluded. Unlike a hallucinated positive finding, such a statement is fluent, typical, and possibly even true, so it is likely to pass unnoticed.

Existing reliability signals do not measure this *evidence sufficiency*. Generation confidence and sampling consistency (C. Wang et al., 2025; Zhang et al., 2025) describe how consistently the generator produces a statement, not whether the image supports it. Hallucination detectors (Hardy et al., 2025; Heiman et al., 2025) and reference-based metrics (Jain et al., 2021; Ostmeier et al., 2024) assess whether content is correct, the latter against reference reports that are unavailable at deployment. Image-quality control (Meng et al., 2022; Selby et al., 2025) yields image-level scores that are not linked to individual statements. None of these signals answers the question a reviewer needs answered: *does the image that the model saw contain the anatomy on which this statement depends?*

We address this question with *source-relative anatomical retention*, the fraction of a predefined lung region in the acquired radiograph that remains in the presented image. Retention is continuous, region-specific, and supervisable at scale from lung masks and controlled presentations, yet it is estimated at inference from the presented image alone. We operationalize it in OA-CXR, an intelligent review-support system that follows the classical expert-system architecture: a knowledge base maps report findings to diagnostic lung regions, a two-stage global–regional network estimates region-specific retention, an inference module converts the estimates into statement flags and budget-constrained report priorities, and an explanation facility reports the finding, region, and estimated retention behind every flag. Because OA-CXR requires only the presented image and the report text, it can audit any generator, including closed-source models (Fig. 1).

We evaluate OA-CXR on five cohorts (9,602 source images; 374,478 anatomical queries) around four research questions: the accuracy of retention estimation (**RQ1**), the contribution of each design component (**RQ2**), review allocation compared with generation-confidence and image-quality

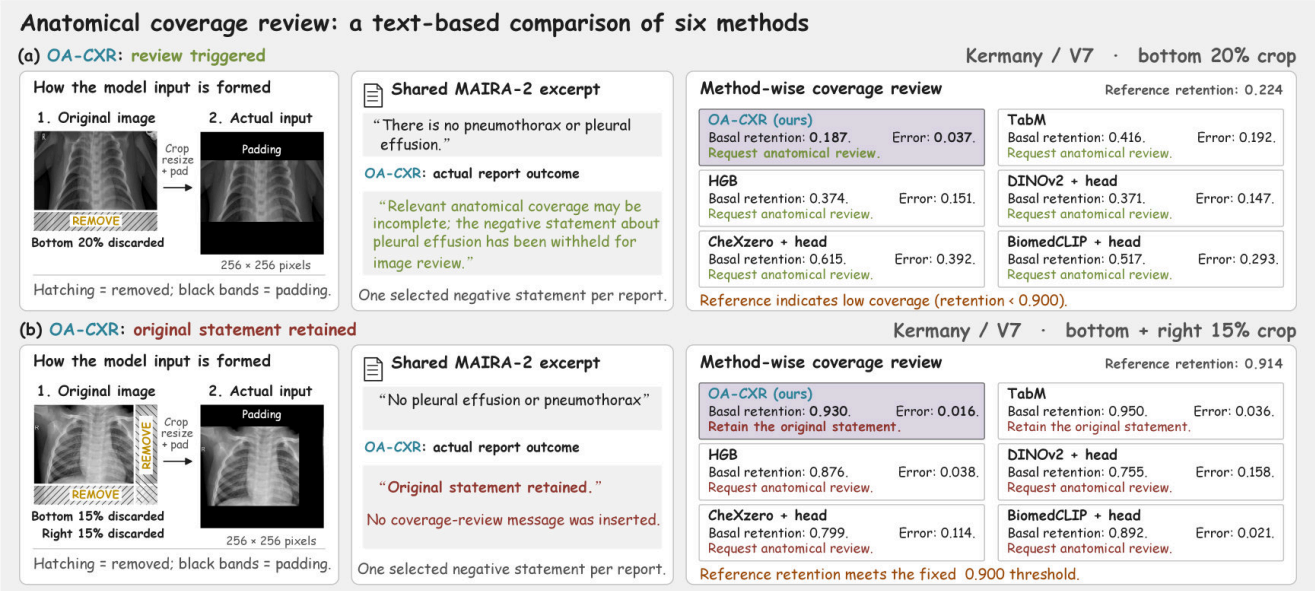


Figure 1: Retention-based review of negative pleural-effusion statements. Each row shows the source and presented images, a report excerpt, the OA-CXR outcome, and basal-retention estimates from six methods. (a) The bottom 20% of the radiograph is removed; the reference basal retention is 0.224, and all six methods request review, with OA-CXR giving the smallest error. (b) The bottom and rightmost 15% are removed; the reference retention is 0.914, and only OA-CXR and TabM correctly retain the statement. Decisions use the fixed review threshold of 0.9; hatching marks removed regions, and black bands indicate padding. Absolute errors are computed before rounding.

signals (**RQ3**), and transfer to an independent clinical cohort (**RQ4**). Our contributions are as follows:

(1) **Problem formulation.** We identify the evidence insufficiency of negative report statements as a failure mode that confidence-based triage cannot detect, and we formalize source-relative anatomical retention as a continuous, statement-linked review target with scalable mask-derived supervision.

(2) **A hybrid knowledge- and data-driven review system.** OA-CXR combines explicit finding-to-region knowledge with a global–regional estimator trained under joint retention and segmentation supervision, producing interpretable statement flags and budget-aware report priorities.

(3) **Empirical evidence.** OA-CXR achieves the lowest error among nine retention regressors on 343,395 public queries. In report review, it attains oracle recall at budgets of up to 30% and captures 91.7% of event-positive reports at a 50% budget, whereas generation confidence performs at chance. Without adaptation, it achieves MAE 0.0281 and AUROC 0.974 on 797 pediatric radiographs from an independent hospital.

2. Related Work

Radiology report generation. Region-guided generation (Tanida et al., 2023) and MAIRA-2 (Bannur et al., 2024) ground report sentences in image regions, and CheXagent, LLaVA-Rad, and MedGemma extend reporting to instruction following (Z. Chen et al., 2024; Sellergren et al., 2025; Zambrano Chaves et al., 2025); L. Huang et al. (2026) review the field. Recent intelligent-systems work has proposed report suggestions (Zeiser et al., 2024), factual serialization (K. Liu et al., 2026), multimodal-LLM refinement (Han et al., 2026), and agent-based decision support for chest radiography (X. Liu et al., 2026), and clinician–VLM studies treat generated reports as drafts that clinicians revise (Tanno et al., 2025). OA-CXR leaves the generator unchanged and audits whether its negative statements are supported by the anatomy it received.

Verifying generated reports. Reference-based measures compare a generated report with a radiologist-written reference through clinical entities (Jain et al., 2021) or LLM-judged errors (Ostmeier et al., 2024), so they are unavailable when a new report is drafted. Reference-free methods estimate hallucination risk from consistency across sampled reports (C. Wang et al., 2025; Zhang et al., 2025) or from the generator's hidden states (Hardy et al., 2025), or they correct measurement statements with dedicated detectors (Heiman et al., 2025); Zhu et al. (2025) review medical hallucination more broadly. These methods ask whether content is *correct* or *consistent*. OA-CXR asks a prior question, namely whether the *evidence* for a statement was available; because a negative statement can be true yet unsupported, this is a distinct review criterion.

Image quality and anatomical representations. NIQE (Mittal et al., 2013), automated chest radiograph quality control (Selby et al., 2025), and deep-learning assessment of field of view and positioning (Meng et al., 2022) operationalize radiographic quality criteria (European Commission, 1996), but their image-level scores are not linked to individual statements. General-purpose encoders such as BioViL-T (Bannur et al., 2023), CheXzero (Tiu et al., 2022), BiomedCLIP (Zhang et al., 2023), and DINOv2 (Oquab et al., 2024) provide transferable features, and HybridGNet (Gaggion et al., 2023) and CheXmask (Gaggion et al., 2024) provide lung segmentation at scale. OA-CXR combines global features

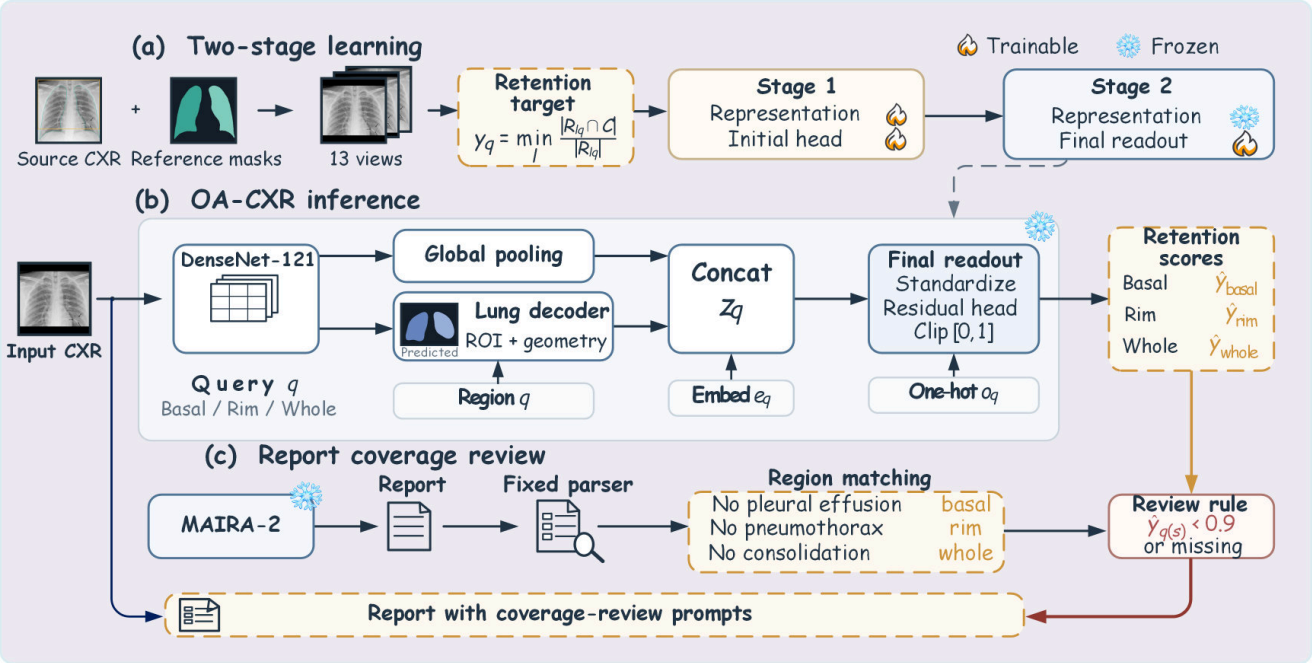


Figure 2: Overview of OA-CXR. (a) Source radiographs and reference lung masks define retention targets for thirteen controlled presentations. Stage 1 learns the anatomical representation and an initial retention head; Stage 2 freezes the representation and fits the final readout. (b) At inference, the presented radiograph and a region query yield a retention score from predicted anatomical regions, without reference masks or the uncropped source image. (c) A fixed parser, driven by the finding-to-region knowledge base, maps eligible negative statements in MAIRA-2 reports to region queries; low or unavailable retention scores trigger coverage-review notices.

with segmentation-guided regional pooling, and we compare it with these encoders.

Selective prediction and deferral. Selective prediction trades coverage against risk (Geifman & El-Yaniv, 2017; Hendrickx et al., 2024), and learning-to-defer methods optimize referral to experts (Mozannar & Sontag, 2020). Deferral to clinicians has improved combined performance in medical imaging (Dvijotham et al., 2023; Leibig et al., 2022), yet a meta-analysis shows that human–AI combinations do not reliably outperform their best component (Vaccaro et al., 2024), underscoring the need for well-targeted deferral signals. Most deferral scores derive from the predictive uncertainty of the audited model; OA-CXR instead targets an explicit, image-grounded criterion.

Summary. Table 1 contrasts the families of review signals. OA-CXR is, to our knowledge, the first system to estimate whether the anatomy required by an individual report statement is present in the image that the generator received, and to use this estimate to allocate clinician review.

3. The OA-CXR System

3.1. Problem Formulation and System Overview

Let I be an acquired source radiograph and $x = T_C(I)$ the presented image obtained by a presentation transform T_C, which crops I to a window C and then resizes and pads it. A report generator receives x and produces a report d. For an eligible statement $s \in d$ about finding $f(s)$, a knowledge base $\mathcal{K}$ returns the diagnostic region query $q(s) = \mathcal{K}(f(s)) \in \mathcal{Q} = \{\text{basal}, \text{rim}, \text{whole}\}$. The evidence available for s is the source-relative retention $y_{q(s)}(I, C)$ defined in Section 3.2. OA-CXR must estimate $\hat{y}_q(x) \approx y_q(I, C)$ from x alone, flag statements whose estimated evidence is insufficient, and rank reports so that a limited review budget reaches the reports most likely to contain such statements.

Figure 2 outlines the workflow, Fig. 3 shows the network, and Algorithm 1 summarizes inference. The system has four modules. The *knowledge base* (Section 3.6, Table 2) encodes where each supported finding manifests radiographically. The *perception module* (Sections 3.2–3.5) is a two-stage global–regional retention estimator: Stage 1 learns image and anatomical representations under joint retention and segmentation supervision, and Stage 2 fits a standardized residual readout to the frozen representations. The *inference module* (Section 3.6) converts retention estimates into statement flags, report edits, and report priorities. The *explanation facility* attaches the finding, region, and estimated retention to every flag, and input-gradient maps of the readout can be inspected on demand (Section 5.4).

3.2. Mask-derived Retention Supervision

Let M_l denote the reference mask of lung $l \in \{1, 2\}$ in the source image I, and let the crop window C be expressed in source-image coordinates. Three reference regions $R_{lq}(I)$ are defined for each lung: the whole-lung mask; a basal region, defined as the inferior quarter of the lung's source-image bounding box; and an inner-rim region, defined as an interior Euclidean-distance band of width $\max(1, 0.02 s_l)$, where s_l is the shorter bounding-box dimension. Fractional boundaries are handled by continuous pixel-area integration. The retention target for query q is

$$y_q(I, C) = \min_{l \in \{1,2\}} \frac{|R_{lq}(I) \cap C|}{|R_{lq}(I)|}. \tag{1}$$

Table 1
Positioning of OA-CXR among review signals for generated chest radiograph reports.

Signal family	Representative methods	What the signal assesses	Granularity	Reference-free at inference	Checks evidence per statement
Reference-based metrics	RadGraph, GREEN	Agreement with a radiologist-written report	Report	✗	✗
Generation uncertainty	Token log probability, entropy, SCUQ, RadFlag	Confidence or consistency of the generator	Sentence / report	✓	✗
Hallucination detection	ReXTrust, FactCheXcker	Correctness of reported findings or measurements	Finding	✓	✗
Image-quality control	NIQE, AutoQC, Meng et al. (2022)	Technical quality and positioning of the image	Image	✓	✗
OA-CXR (ours)	This work	Presence of the anatomy required by each statement	Statement / region	✓	✓

Algorithm 1 OA-CXR review of one generated report.

Require: Presented image x; generated report d; knowledge base $\mathcal{K}$; review threshold τ_{review}
Ensure: Edited report $\tilde{d}$; report priority ρ_d; explanations E

$(g, A, P) \leftarrow \textsc{EncodeDecode}(x)$ ▷ one pass for all queries
for $q \in \{\text{basal}, \text{rim}, \text{whole}\}$ **do**
  $z_q \leftarrow \textsc{Fuse}(g, A, P, q)$; $\hat{y}_q \leftarrow \textsc{Readout}(z_q)$
$S_d \leftarrow \textsc{Parse}(d, \mathcal{K})$ ▷ eligible statements and queries
$\tilde{d} \leftarrow d$; $E \leftarrow \emptyset$
for $s \in S_d$ **do**
  $\rho_s \leftarrow 1 - \hat{y}_{q(s)}$
  if $\hat{y}_{q(s)} < \tau_{\text{review}}$ **or** $\hat{y}_{q(s)}$ unavailable **then**
    $\tilde{d} \leftarrow \textsc{InsertNotice}(\tilde{d}, s)$
    $E \leftarrow E \cup \{(f(s), q(s), \hat{y}_{q(s)})\}$
$\rho_d \leftarrow \max_{s \in S_d} \rho_s$ ▷ rank reports by ρ_d
return $\tilde{d}$, ρ_d, E

The minimum makes the target sensitive to anatomical loss on either side. Targets are computed in source-image coordinates before resizing and padding; an uncropped source has retention one, and transformations that preserve the retained anatomical support leave the target unchanged. Reference masks are needed only to construct training targets; at inference, OA-CXR estimates the target from the presented image.

Each source image yields thirteen presentations: the original, eight single-edge crops that remove 10% or 20% of one edge, and four corner crops that remove 15% from each of two adjacent edges. Images are resized with preserved aspect ratio and zero-padded to 256×256, using bilinear interpolation for intensities and nearest-neighbor interpolation for masks, and intensities are mapped to the encoder range $[-1024, 1024]$. Only the resulting image and the query are passed to the retention model.

3.3. Global–Regional Anatomical Representation

A DenseNet-121 encoder (G. Huang et al., 2017) is initialized with CheXpert-trained TorchXRayVision weights (Cohen et al., 2022; Irvin et al., 2019). A $1 \times 256 \times 256$ input produces $F \in \mathbb{R}^{1024\times8\times8}$, and global average pooling yields $g \in \mathbb{R}^{1024}$. The anatomy decoder projects F to 64 channels and applies two bilinear upsampling blocks with 3×3 convolutions, four-group normalization, and SiLU activation, producing $A \in \mathbb{R}^{64\times32\times32}$. A separate 1×1 convolution followed by a sigmoid predicts two lung probability maps, $P \in [0, 1]^{2\times32\times32}$.

Regional supports are constructed from the predicted lung probabilities. A decoder-grid location belongs to lung l when $P_l(p) \geq 0.5$, and its probability is retained as a pooling weight. The basal support is the inferior quarter of the predicted lung bounding box. The inner-rim support is obtained by square-kernel erosion with radius $\max(1, \lceil 0.02\widetilde{s}_l \rceil)$, where $\widetilde{s}_l$ is the shorter predicted bounding-box dimension on the decoder grid. These supports define pooling regions on the decoder grid, whereas the reference regions in Eq. (1) define supervision in source-image coordinates.

Let W_{lq} denote the predicted weight map for lung l and query q, and define $W_q(p) = \max_l W_{lq}(p)$. For a weight map W on the decoder grid Ω,

$$\Pi(A, W) = \frac{\sum_{p\in\Omega} W(p)A(:, p)}{\max\left(\sum_{p\in\Omega} W(p), \epsilon\right)}, \qquad \epsilon = 10^{-6}. \quad (2)$$

The query-region vector is $r_q = \Pi(A, W_q) \in \mathbb{R}^{64}$, and the bilateral whole-lung context vectors are $b_l = \Pi(A, W_{l,\text{whole}}) \in \mathbb{R}^{64}$; empty supports yield zero vectors. The fused representation concatenates global, regional, bilateral, geometric, and query features:

$$z_q = [g; r_q; b_1; b_2; d_q; e_q] \in \mathbb{R}^{1250}, \quad (3)$$

where $d_q \in \mathbb{R}^{18}$ comprises 14 probability-map statistics, three normalized support masses, and a regional-branch-use flag, and $e_q \in \mathbb{R}^{16}$ is a learned query embedding, giving $1024 + 64 + 128 + 18 + 16 = 1250$ dimensions (Fig. 3b). The global vector can encode context indicative of truncation, such as the position of the diaphragm and chest wall relative to the image border, whereas the regional vectors describe the specific support queried by a statement.

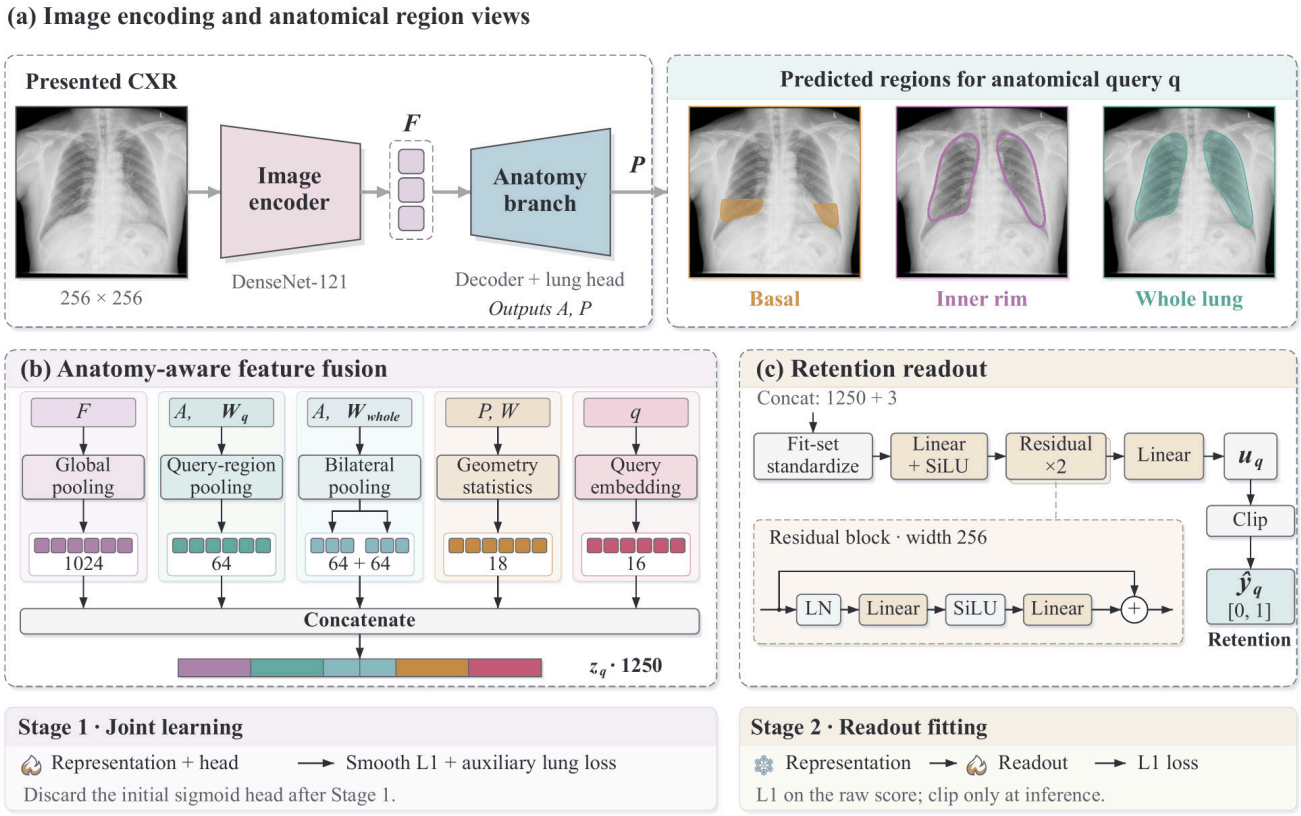


Figure 3: Architecture of the OA-CXR perception module. (a) A DenseNet-121 encoder extracts features F, and the anatomy branch outputs decoder features A and lung probability maps P, from which the basal, inner-rim, and whole-lung supports are derived. (b) Global, query-region, bilateral whole-lung, geometric, and query-embedding features are concatenated into the 1,250-dimensional representation z_q. (c) The readout appends a one-hot query code, applies fit-set standardization, a linear SiLU projection, and two residual blocks of width 256, and clips the raw score u_q to $[0, 1]$ at inference. Bottom: Stage 1 jointly trains the representation and an initial head with Smooth L1 and auxiliary lung losses; Stage 2 freezes the representation and fits the readout with an L1 loss on u_q.

3.4. Stage 1: Joint Representation Learning

An initial shared head maps z_q through $1250 \to 128 \to 1$ with SiLU activation and a sigmoid output. The encoder, anatomy decoder, lung head, query embedding, and initial retention head are jointly optimized with

$$\mathcal{L}_{\text{repr}} = \mathcal{L}_{\text{ret}} + 0.25\left(\mathcal{L}_{\text{BCE}} + 0.5\mathcal{L}_{\text{Dice}}\right), \tag{4}$$

where $\mathcal{L}_{\text{ret}}$ is the Smooth L1 loss with $\beta = 0.1$, averaged over the three queries, and the segmentation losses compare transformed reference masks with the upsampled lung predictions. Segmentation supervision forces the decoder to localize the lungs, so the regional supports in Eq. (2) follow anatomy rather than image position.

During training, the query-region vector, bilateral context, and associated geometric descriptors are jointly dropped with probability 0.2, while global features, query embeddings, and segmentation supervision remain active; this regional-branch dropout discourages the head from relying exclusively on the regional pathway. Encoder BatchNorm running statistics are fixed, whereas convolutional weights and affine parameters are trainable. The initial retention head is used only in Stage 1 and is discarded after checkpoint selection.

3.5. Stage 2: Standardized Residual Readout

In Stage 2, the encoder, anatomy decoder, lung head, and query embedding are frozen, and regional dropout is disabled. A three-dimensional one-hot query code o_q is appended to Eq. (3):

$$v_q = [z_q; o_q] \in \mathbb{R}^{1253}, \qquad \tilde{v}_q = (v_q - \mu_{\text{fit}}) \oslash s_{\text{fit}}. \tag{5}$$

The featurewise mean and population standard deviation are estimated from fit-set features and held fixed during development, evaluation, and inference; standard deviations below 10^{-6} are replaced by one. Standardization places the heterogeneous global, regional, geometric, and query features on a common scale before they enter the readout.

As shown in Fig. 3(c), a linear projection with SiLU activation maps the standardized input to $h_0 \in \mathbb{R}^{256}$. Two residual blocks then compute

$$h_{k+1} = h_k + W_{k2}\,\text{SiLU}\big(W_{k1}\,\text{LN}(h_k) + b_{k1}\big) + b_{k2}, \quad k \in \{0, 1\}, \tag{6}$$

followed by the scalar output $u_q = w_{\text{out}}^{\top} h_2 + b_{\text{out}}$. Only the readout parameters are optimized, using

$$\mathcal{L}_{\text{readout}} = \frac{1}{|\mathcal{D}_{\text{fit}}||\mathcal{Q}|} \sum_{x \in \mathcal{D}_{\text{fit}}} \sum_{q \in \mathcal{Q}} |u_q(x) - y_q(x)|, \tag{7}$$

where $\mathcal{D}_{\text{fit}}$ is the set of presented fit images. The readout is trained on unconstrained outputs, and the reported prediction is $\hat{y}_q = \min(1, \max(0, u_q))$. The readout checkpoint is selected by development-set MAE, and a single encoder–decoder pass provides the shared features for all three queries.

Table 2
Finding-to-region knowledge base used by the parser.

Finding	Query	Radiological rationale
Pleural effusion	Basal	Fluid first collects in the dependent costophrenic recesses
Pneumothorax	Inner rim	The visceral pleural line is sought along the lung periphery
Consolidation	Whole lung	Air-space opacification may involve any lobe or segment

3.6. Knowledge-Guided Statement Review and Report Prioritization

Knowledge base. The knowledge base encodes, for each supported finding, the lung region whose visibility is required to support a negative statement about it (Table 2). The mappings follow standard radiological signs and quality criteria (European Commission, 1996): on erect radiographs, small pleural effusions first blunt the costophrenic angles at the lung bases; a pneumothorax is recognized by a visceral pleural line separated from the chest wall along the lung periphery; and consolidation can involve any part of the lung parenchyma. Because the knowledge base is explicit, each flag can be traced to a human-readable rule, and new findings can be added by specifying their diagnostic regions.

Statement flags. Reports are generated with frozen MAIRA-2 (Bannur et al., 2024). A fixed rule-based parser identifies explicit negative statements about the three findings and maps them to their queries. For an eligible statement s,

$$e_s = \mathbf{1}\left[y_{q(s)} < \tau_{\text{event}}\right], \qquad \rho_s = 1 - \hat{y}_{q(s)}, \tag{8}$$

with $\tau_{\text{event}} = 0.9$. The reference event e_s thus marks a statement for which more than 10% of the required region is missing. A statement is flagged when $\hat{y}_{q(s)} < \tau_{\text{review}} = 0.9$ or its score is unavailable. A deterministic editing policy withholds or restructures flagged statements and inserts a coverage-review notice (Fig. 1a).

Report prioritization. For a report d with eligible statements S_d,

$$\rho_d = \max_{s \in S_d} \rho_s, \qquad e_d = \max_{s \in S_d} e_s. \tag{9}$$

Reports with eligible statements are ranked by decreasing ρ_d, and review budgets of 5%, 10%, 20%, 30%, and 50% of this eligible subset are evaluated. Exact ties receive equal fractional selection probabilities at the budget boundary. Unavailable scores are routed to mandatory review; none occurred in the fixed-workload comparison.

3.7. Retention–Confidence Combination

To test whether generation confidence adds information beyond retention, we fit a logistic model on a separate calibration partition. For statement s in report d, it combines the standardized retention deficit $a_s = 1 - \hat{y}_{q(s)}$ and the report's negative teacher-forced mean token log probability ℓ_d:

$$\rho_s^{\text{comb}} = \sigma\left(b + \beta_1 \frac{a_s - \mu_1}{s_1} + \beta_2 \frac{\ell_d - \mu_2}{s_2}\right), \tag{10}$$

fitted to e_s with group-balanced weights, L2 regularization ($C = 1$), and LBFGS optimization (seed 17); standardization statistics come from calibration data only, and all parameters are then held fixed. A whole-lung-only control assigns the whole-lung score to every eligible statement while preserving its finding-specific event label, isolating the contribution of the knowledge base.

4. Experimental Setup

4.1. Datasets

We evaluate OA-CXR on five chest radiograph cohorts: COVID-QU-Ex (Tahir et al., 2021a, 2021b), Kermany/V7 (Kermany et al., 2018; V7 Labs, n.d.), NIH/CheXmask (Gaggion et al., 2024; X. Wang et al., 2017), Shenzhen (Jaeger et al., 2014; Stirenko et al., 2018), and an external pediatric cohort from *[institution withheld for double-anonymized review]*. COVID-QU-Ex and Kermany/V7 are used for model fitting and checkpoint selection, whereas NIH/CheXmask, Shenzhen, and the clinical cohort are used only for evaluation. Lung masks define the reference regions in Eq. (1): COVID-QU-Ex provides human–machine collaborative annotations (Tahir et al., 2021a), CheXmask provides HybridGNet-derived masks (Gaggion et al., 2023, 2024) through PhysioNet (Gaggion et al., 2025; Pollard et al., 2026), and the clinical cohort was annotated manually by clinicians.

During construction of the public cohorts, image-level duplicate screening with exact and perceptual fingerprints excluded 4,541 duplicate source images. COVID-QU-Ex was partitioned by deterministic hashing into fit, development, calibration, and held-out evaluation sets. Kermany images were grouped by filename family before partitioning, and families containing official test images were assigned to evaluation. All presentations derived from a source image remained in the same partition.

The clinical cohort initially comprised 800 pediatric chest radiographs, primarily from cases related to respiratory infection. After three exact duplicates were removed, 797 images remained, with patient ages from 10 days to 13 years (median, 3 years 7 months). The images formed 791 identity groups for statistical resampling and were reserved for external evaluation. Clinicians annotated bilateral lung regions. The retrospective study was approved by the Institutional Review Board (IRB)/Ethics Committee of *[institution withheld for double-anonymized review]* (*[approval identifier and date withheld for double-anonymized review]*), which waived the requirement for informed consent for the retrospective use of de-identified data.

Each source image was rendered with the thirteen-presentation protocol of Section 3.2, and basal, inner-rim, and whole-lung queries were generated for each presentation (Table 3). The clinical cohort yielded 10,361 presentations and 31,083 queries, of which 8,589 (27.63%) had reference retention below 0.9.

4.2. Implementation and Comparison Methods

OA-CXR is implemented in Python 3.10 and PyTorch 2.6, with TorchXRayVision 1.5.4 for encoder initialization. Both stages use AdamW and seed 17. Stage 1 runs for 13 epochs with batch size 256, learning rate 10^{-4}, weight decay 10^{-4}, BF16 mixed precision, and regional-branch dropout

Table 3
Dataset partitions and standard evaluation sizes after duplicate screening. Counts refer to source images before generation of thirteen presentations and three queries per presentation. Resampling groups are filename families for Kermany/V7, identity groups for the clinical cohort, and source images for the remaining datasets.

	Source images				Standard evaluation		
Source	Fit	Dev.	Calib.	Evaluation	Presentations	Queries	Groups
COVID-QU-Ex	16,192	2,255	2,289	1,855	24,115	72,345	1,855
Kermany/V7	2,894	354	362	937	12,181	36,543	360
NIH/CheXmask	0	0	0	5,821	75,673	227,019	5,821
Shenzhen	0	0	0	192	2,496	7,488	192
Clinical cohort	0	0	0	797	10,361	31,083	791
Total	19,086	2,609	2,651	9,602	124,826	374,478	9,019

0.2. A deterministic cyclic schedule presents every source image in all 13 conditions over training, giving 975 optimizer updates; development performance selects epoch 12. Stage 2 trains the readout for 30 epochs on all query instances, with batch size 2,048, FP32 precision, weight decay 10^{-4}, and cosine learning-rate decay from 10^{-3} to 10^{-5}; development MAE selects epoch 27.

Retention regressors. The eight regression comparators comprise five alternative readouts of the same frozen OA-CXR features, namely histogram gradient boosting (HGB) and Ridge regression (Pedregosa et al., 2011), XGBoost (T. Chen & Guestrin, 2016), CatBoost (Prokhorenkova et al., 2018), and TabM (Gorishniy et al., 2025), and three frozen pretrained encoders with fitted readouts: BiomedCLIP (Zhang et al., 2023), CheXzero (Tiu et al., 2022), and DINOv2 (Oquab et al., 2024). The shared-feature readouts isolate the contribution of the readout, whereas the foundation-model encoders test whether general-purpose features suffice.

Review signals. Image-based ranking signals are CheXpert-only (Irvin et al., 2019), AutoQC CXR-NIQE (Mittal et al., 2013; Selby et al., 2025), and BioViL-T (Bannur et al., 2023). Report-level signals are mean token log probability, sentence-level log probability, negative token entropy, and SCUQ-RadGraph (Jain et al., 2021; C. Wang et al., 2025). Quality, confidence, and uncertainty scores have no retention scale and are therefore assessed with ranking metrics only.

Structural controls. The global DenseNet control uses the same initialization and training schedule but removes the anatomy decoder, segmentation objective, and regional pooling. The predicted-mask geometry control fits an HGB regressor to 75 geometric and query features derived from the frozen OA-CXR segmentation output, testing whether explicit geometry alone suffices (denoted Mask geometry + HGB in the tables).

External clinical evaluation uses frozen models without clinical-cohort training, checkpoint selection, or score-direction adjustment. Its fourteen configurations comprise the eight regression comparators, three image-based ranking signals, two structural controls, and OA-CXR.

4.3. Metrics and Statistical Analysis

Retention regression is evaluated by mean absolute error (MAE) and root mean squared error (RMSE). Macro MAE and macro RMSE are arithmetic means of the source-specific metrics, with each of the four public datasets weighted equally. For the low-retention event $y_q < 0.9$, we report AUROC, average precision (AP), and the area under the risk–coverage curve (AURC), where risk is the proportion of low-retention observations among those not withheld and coverage is the fraction retained. Fixed-workload review is evaluated by event recall and by the event proportion among unreviewed eligible reports. We additionally report two analytical references: random allocation, whose expected recall equals the budget, and an oracle that reviews event-positive reports first. Comparisons use common valid sets, with missing scores and denominators reported separately.

For the public datasets, 95% confidence intervals are estimated from 1,000 paired grouped bootstrap replicates that retain all presentations and queries within each resampled group (Table 3); all methods use identical resamples, and pooled workload analyses are additionally stratified by dataset. Because all models are fitted with seed 17, these intervals quantify evaluation-sample uncertainty conditional on the fitted models. Ablation comparisons use unadjusted paired intervals.

Clinical-cohort metrics are reported separately from the public macro averages. Pointwise 95% percentile intervals are estimated from 1,000 paired bootstrap resamples of the 791 identity groups. For the comparisons in Fig. 6(e–l), significance is assessed with approximate two-sided paired delete-one-group jackknife t-tests with 790 degrees of freedom, and Holm adjustment covers all 80 comparisons: ten comparators versus OA-CXR for overall MAE and AUROC and for the three region-specific MAEs and RMSEs.

5. Results

5.1. RQ1: Accuracy of Retention Estimation

Across the 343,395 public queries, OA-CXR achieves the lowest MAE on all four datasets and the lowest macro RMSE (Table 4). Its macro MAE of 0.0296 is 3.69% lower than that of TabM, the strongest alternative readout of the same features, 23.1% lower than that of HGB, and 23.9% lower than that of DINOv2, the strongest frozen foundation-model encoder. Paired source-specific MAE intervals favor OA-CXR over TabM on all four datasets (Fig. 4d). Two observations follow. First, the learned representation is the main source of accuracy: the two strongest readouts of the OA-CXR features, TabM and HGB, both outperform all three frozen foundation-model encoders, whereas a linear readout (Ridge) falls behind DINOv2 and CheXzero, indicating that retention is encoded nonlinearly in the representation. Second,

Table 4
Anatomical-retention regression on the four public evaluation sources (343,395 queries). Values are $\times 10^2$. Source columns report MAE; macro MAE and macro RMSE weight the four sources equally. Lower is better; bold indicates the best result.

Group	Method	COVID-QU-Ex	Kermany/V7	NIH/CheXmask	Shenzhen	Macro MAE	Macro RMSE
Alternative readouts of frozen OA-CXR features	HGB	3.633	4.708	3.350	3.727	3.855	7.089
	XGBoost	3.828	4.878	3.548	3.873	4.032	7.313
	CatBoost	3.815	4.909	3.521	3.892	4.034	7.289
	Ridge	4.421	5.266	4.110	4.326	4.531	7.665
	TabM	2.823	3.748	2.655	3.080	3.077	6.475
Frozen foundation-model encoders + fitted head	BiomedCLIP	4.910	5.454	4.673	4.522	4.890	9.341
	CheXzero	3.988	4.805	3.728	4.275	4.199	8.414
	DINOv2	3.881	4.758	3.634	3.293	3.892	7.634
	OA-CXR (ours)	**2.695**	**3.683**	**2.534**	**2.940**	**2.963**	**6.265**

Table 5
Ablations of the fixed representation and of the full training pipeline. MAE values are $\times 10^2$. Dev. denotes the readout selected by development MAE and Last its epoch-30 counterpart. Relative increases are computed against the Full control of each panel from unrounded values. Paired 95% confidence intervals are reported for $\mathrm{MAE}_{\mathrm{Full}} - \mathrm{MAE}_{\mathrm{variant}}$ at the Dev. endpoint in units of 10^{-3}; intervals entirely below zero favor Full. Bold indicates the lowest macro MAE in each panel.

Condition	Dev. macro MAE	Increase (%)	Last macro MAE	95% CI ($\times 10^{-3}$)
(a) Fixed-representation and readout ablations				
Without regional inputs	3.089	+3.67	3.089	[−1.3312, −0.8522]
Without global inputs	3.165	+6.23	3.170	[−2.1964, −1.5587]
Without geometric descriptors	2.994	+0.46	3.010	[−0.2942, +0.0228]
Without explicit query encoding	2.989	+0.29	3.007	[−0.2644, +0.0878]
Without standardization	3.173	+6.49	3.173	[−2.3128, −1.5914]
Without residual connections	2.984	+0.15	2.988	[−0.3264, +0.2186]
Smooth L1 instead of L1	3.006	+0.89	3.036	[−0.4265, −0.0981]
Sigmoid output	2.992	+0.41	2.998	[−0.3755, +0.1571]
OA-CXR (Full)	**2.980**	—	**2.987**	—
(b) Full-pipeline ablations				
Without segmentation loss	3.226	+8.15	3.243	[−2.8432, −2.0532]
Without regional features	3.048	+2.19	3.064	[−0.9782, −0.3446]
Without regional-branch dropout	3.007	+0.81	3.007	[−0.4872, −0.0033]
Frozen image encoder	3.632	+21.76	3.641	[−7.0562, −5.9522]
OA-CXR (Full)	**2.983**	—	**2.983**	—

the residual readout extracts this information more effectively than any alternative readout of the same features.

The prediction histograms in Fig. 4(a) show that OA-CXR predictions concentrate most tightly around the identity line; points above and below the diagonal correspond to over- and underestimation of the reference retention. For detecting the low-retention event, OA-CXR achieves the highest AUROC on all four datasets (0.9705–0.9836) and the highest AP on COVID-QU-Ex, Kermany/V7, and NIH/CheXmask (Fig. 4b). Image-based signals that were not designed for this task, namely CheXpert-only, AutoQC CXR-NIQE, and BioViL-T, reach AUROC values of only 0.4170–0.5462, confirming that generic image quality or semantics does not reveal which anatomy has been lost. The risk–coverage curves (Fig. 4c) translate this ranking into selective behavior: withholding the lowest-scored queries keeps the low-retention fraction small among the retained queries until coverage approaches 100%.

5.2. RQ2: Contributions of System Components

We assess component contributions with eight fixed-representation and four full-pipeline variants, each compared with a retrained Full control (Table 5, Fig. 5). Fixed-representation variants retrain only the readout, whereas full-pipeline variants repeat both stages; all use the same partitions, seed, and schedules. The two Full controls reach macro MAEs of 0.02980 and 0.02983, which closely reproduce the main model.

Primary contributors. Four design choices account for most of the accuracy. Freezing the encoder increases macro MAE by 21.76%, showing that task-specific adaptation of the image encoder is essential. Removing segmentation supervision increases MAE by 8.15%, and the effect is also visible in localization: on the four public datasets, lung Dice falls from 0.908–0.931 for Full to 0.178–0.231, whereas the other full-pipeline variants retain Dice of 0.861–0.936 (Fig. 5f–i). Removing standardization or global inputs from the readout increases MAE by 6.49% and 6.23%, respectively, and removing regional inputs or regional features increases it by 3.67% and 2.19%. All corresponding paired intervals

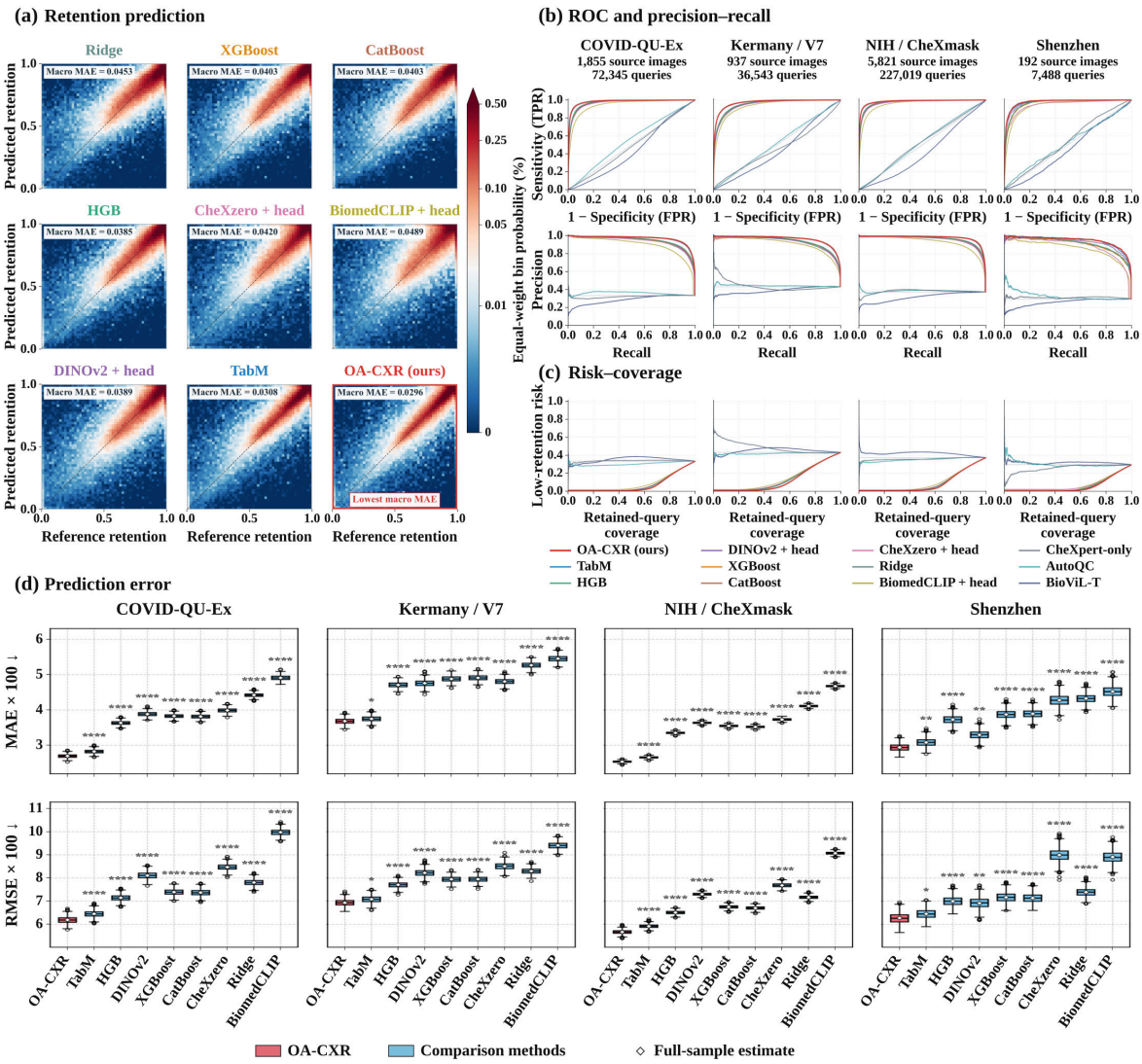


Figure 4: Comparison on four public datasets (8,805 radiographs; 343,395 queries). **(a)** Reference–prediction agreement. **(b)** ROC and precision–recall curves for the low-retention event $y_q < 0.9$. **(c)** Risk–coverage curves. **(d)** MAE and RMSE (×100) from 1,000 paired grouped bootstrap replicates; diamonds denote full-sample estimates. Adjusted significance: $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, $****P < 0.0001$; ns, $P \geq 0.05$.

exclude zero, indicating that global and regional information are complementary rather than redundant.

Secondary refinements. Using L1 rather than Smooth L1 in the readout (0.89%) and regional-branch dropout (0.81%) yield small but statistically supported gains. Removing geometric descriptors, explicit query encoding, or residual connections, or using a sigmoid output, changes MAE by 0.15–0.46% with intervals that include zero; these components act as low-cost refinements rather than essential elements. Explicit query encoding is largely redundant because regional pooling is already query-dependent.

Effect on review decisions. Regression gains carry over to review. Every full-pipeline variant both misses more low-retention statements and flags more above-threshold statements than Full (Fig. 5e): among the 784 eligible statements, Full leaves 29 low-retention statements unflagged and raises 14 unnecessary flags, compared with 33–39 and 18–22 for the variants.

Structural controls. Anatomical modeling improves on global image features alone (Table 6). OA-CXR reduces macro MAE from 0.0323 for the global DenseNet control, trained on the same schedule, to 0.0296, an 8.38% reduction, with paired intervals favoring OA-CXR on all four datasets. Conversely, explicit geometry alone is insufficient: an HGB regressor on 75 geometric and query descriptors from the same segmentation reaches only 0.0622. Accurate retention estimation therefore requires learned image features and anatomical structure together.

Table 6
Structural controls on the same 343,395 queries. MAE values are $\times 10^2$; macro MAE weights the four sources equally. The geometry control uses the frozen OA-CXR segmentation output. Lower is better; bold indicates the best result.

Method	COVID-QU-Ex	Kermany /V7	NIH/ CheXmask	Shenzhen	Macro
Global DenseNet	2.977	4.000	2.736	3.224	3.234
Mask geometry + HGB	6.056	6.735	5.676	6.411	6.219
OA-CXR	**2.695**	**3.683**	**2.534**	**2.940**	**2.963**

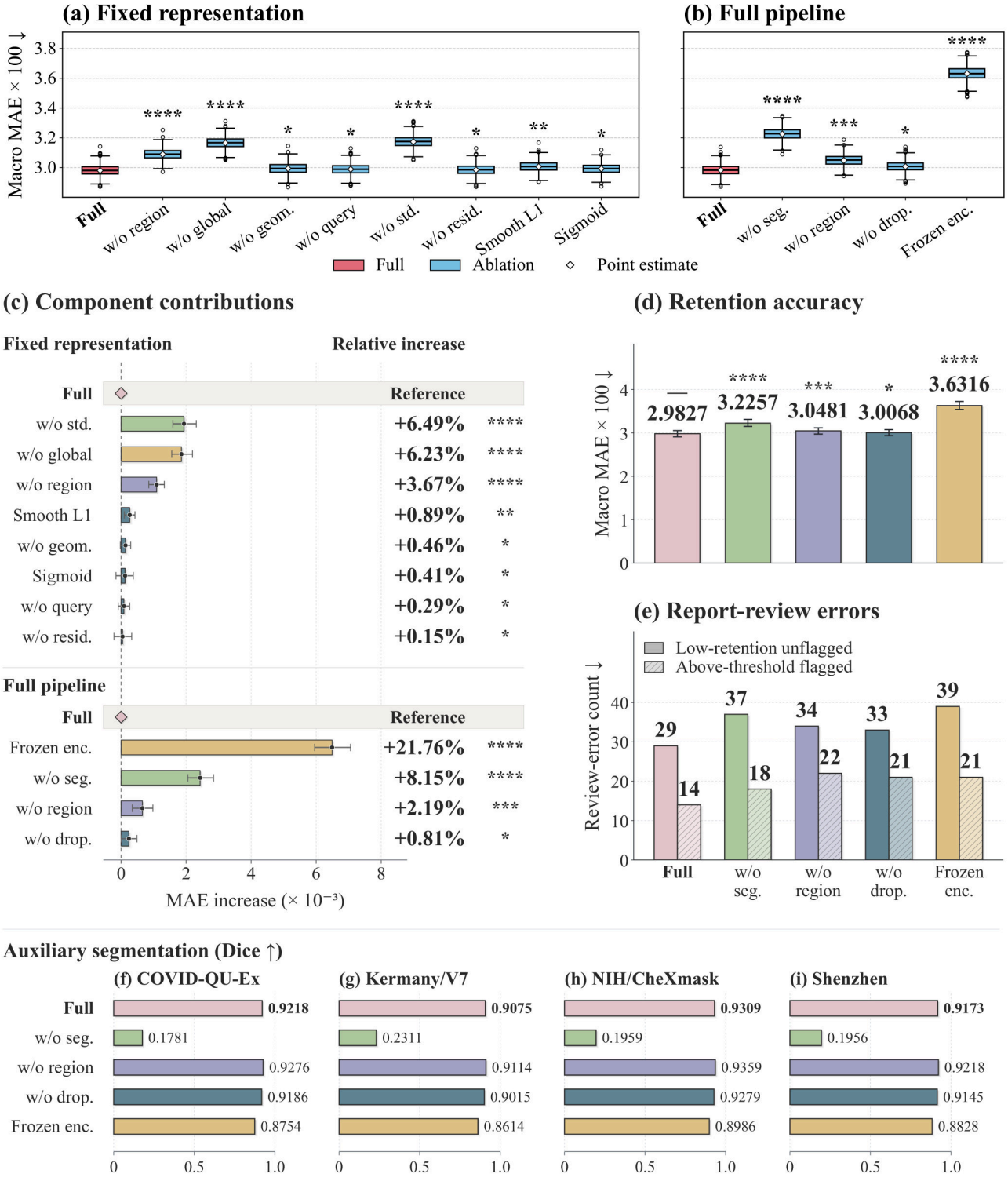


Figure 5: Ablations of OA-CXR. **(a,b)** Bootstrap distributions of macro MAE for fixed-representation and full-pipeline variants; diamonds mark full-sample estimates. **(c)** MAE increase over the separately retrained Full control of each family, with relative increases. **(d)** Full-pipeline macro MAE. **(e)** Review errors among the 784 eligible statements at the threshold of 0.9. **(f–i)** Lung-segmentation Dice of the full-pipeline variants on the four public datasets. Results use 1,000 paired grouped bootstrap replicates; error bars show 95% CIs. Adjusted significance versus Full: $*P < 0.05$, $**P < 0.01$, $***P < 0.001$, $****P < 0.0001$. Abbreviations: std., standardization; geom., geometric descriptors; resid., residual connections; seg., segmentation loss; drop., regional-branch dropout; enc., encoder; region, regional inputs (fixed representation) or regional features (full pipeline).

Table 7
Statement-level detection and selective review of low anatomical retention. Each entry reports AUROC↑ / AP↑ / AURC↓. Panel (a) compares image-based and report-uncertainty signals; panel (b) compares retention regressors (tree-based and linear readouts of the OA-CXR features are compared at the query level in Table 4). Comparisons use the statements available to all methods in each panel; bold indicates the best point estimate.

(a) Image-based and report-uncertainty signals: 741 common statements			
Method	COVID-QU-Ex ($n = 136$)	Kermany/V7 ($n = 432$)	NIH/CheXmask ($n = 173$)
CheXpert-only	0.4635 / 0.3607 / 0.3763	0.4716 / 0.5322 / 0.5631	0.4960 / 0.5320 / 0.4916
AutoQC CXR-NIQE	0.5333 / 0.4607 / 0.3640	0.6081 / 0.5979 / 0.4521	0.5112 / 0.4904 / 0.4872
BioViL-T	0.5483 / 0.4597 / 0.3423	0.4996 / 0.5124 / 0.5293	0.4559 / 0.4595 / 0.5048
Report log probability	0.4544 / 0.3690 / 0.4516	0.5386 / 0.5921 / 0.5257	0.3565 / 0.4185 / 0.6359
Sentence log probability	0.3867 / 0.3587 / 0.5119	0.5179 / 0.5606 / 0.5548	0.5210 / 0.5193 / 0.4627
Negative token entropy	0.3826 / 0.3385 / 0.4913	0.5040 / 0.5499 / 0.5645	0.5321 / 0.5226 / 0.4452
SCUQ-RadGraph	0.4732 / 0.3832 / 0.4151	0.5406 / 0.5451 / 0.4897	0.3999 / 0.4469 / 0.5864
OA-CXR (ours)	**0.9952** / **0.9915** / **0.0915**	**0.9810** / **0.9846** / **0.1797**	**0.9921** / **0.9927** / **0.1570**
(b) Retention regressors: 784 common statements			
Method	COVID-QU-Ex ($n = 141$)	Kermany/V7 ($n = 464$)	NIH/CheXmask ($n = 179$)
TabM	0.9889 / 0.9820 / 0.1014	0.9800 / 0.9801 / **0.1703**	0.9899 / 0.9888 / 0.1463
DINOv2 + head	0.9524 / 0.9499 / 0.1288	0.9673 / 0.9647 / 0.1785	0.9729 / 0.9722 / 0.1549
CheXzero + head	0.9519 / 0.9403 / 0.1226	0.9658 / 0.9663 / 0.1772	0.9527 / 0.9104 / 0.1613
BiomedCLIP + head	0.9263 / 0.8771 / 0.1278	0.9403 / 0.9394 / 0.1904	0.9577 / 0.9472 / 0.1604
OA-CXR (ours)	**0.9912** / **0.9869** / **0.1006**	**0.9803** / **0.9830** / 0.1742	**0.9926** / **0.9927** / **0.1453**

Table 8
Review decisions at the fixed threshold $\hat{y}_q < 0.9$ for the 784 eligible statements, of which 379 are low-retention events. "Missed" denotes $y_q < 0.9$ with $\hat{y}_q \geq 0.9$; "False flags" denotes $y_q \geq 0.9$ with $\hat{y}_q < 0.9$. Sensitivity and specificity (%) are derived from these counts. Bold indicates the best value.

Method	Missed	False flags	Total flagged	Sens.	Spec.
TabM	35	**17**	361	90.8	**95.8**
DINOv2 + head	35	37	381	90.8	90.9
CheXzero + head	43	38	374	88.7	90.6
BiomedCLIP + head	50	60	389	86.8	85.2
OA-CXR (ours)	**29**	**17**	367	**92.3**	**95.8**

5.3. RQ3: Statement-Level Detection and Report Review

Statement-level detection. The 6,656 generated reports contain 784 parser-eligible negative statements, of which 379 (48.3%) are low-retention events. Table 7 compares methods on the statements available to all methods in each panel: 741 in panel (a), after excluding 43 statements without SCUQ scores, and all 784 in panel (b). Shenzhen contributes no eligible statements. OA-CXR achieves the highest AUROC and AP on all three evaluable datasets in both panels and the lowest AURC in every comparison except Kermany/V7 in panel (b), where TabM is slightly lower (0.170 versus 0.174). In contrast, every generation-confidence and uncertainty signal performs close to or below chance (AUROC 0.36–0.54), as do the image-quality and image–text signals (0.46–0.61). These signals measure properties of the text or of the image as a whole, not whether the anatomy required by a statement was visible.

Fixed-threshold decisions. At the review threshold of 0.9, OA-CXR misses 29 low-retention statements and raises 17 false flags, giving a sensitivity of 92.3% and a specificity of 95.8% (Table 8). TabM matches this specificity but misses six more events, and the frozen foundation-model encoders miss 35–50 events while raising 37–60 false flags. Figure 1 illustrates both sides of the threshold: in panel (a), all methods request review of a statement whose basal region is largely missing, and OA-CXR gives the closest estimate (0.187 versus a reference of 0.224); in panel (b), only OA-CXR and TabM correctly retain a statement whose reference retention (0.914) lies just above the threshold.

Table 9
Report-level recall (%) of event-positive reports under fixed review workloads. Budgets are fractions of the 445 eligible reports; recall is computed over the 235 reports containing at least one low-retention event. Random allocation (expected recall equal to the budget) and the oracle (event-positive reports reviewed first) are analytical references. The whole-lung control changes the ranking score but keeps the finding-specific event definition. Bold indicates the best non-oracle value, including ties.

	Review budget				
Method	5%	10%	20%	30%	50%
Random allocation	5.0	10.0	20.0	30.0	50.0
Token log probability	4.3	10.4	20.9	28.9	50.0
OA-CXR, whole lung	**9.5**	**18.9**	37.4	56.0	89.1
OA-CXR, finding-specific	**9.5**	**18.9**	**37.9**	**56.8**	**91.7**
Oracle (upper bound)	*9.5*	*18.9*	*37.9*	*56.8*	*94.7*

Review under fixed workloads. The fixed-workload analysis covers the 6,656 MAIRA-2 reports, of which 445 contain eligible statements and 235 (52.8%) contain at least one low-retention event. We compare finding-specific OA-CXR, the whole-lung-only control, mean token log probability, and the retention–confidence combination calibrated on 761 eligible statements from 152 source groups in the separate calibration partition; all methods score every eligible statement. Confidence intervals are estimated from 1,000 paired source-stratified grouped bootstrap replicates.

Table 10
External validation on the independent pediatric cohort (797 radiographs; 31,083 queries) against clinician-annotated references. All metrics are $\times 10^2$, and every configuration scored all queries. (a) Retention regressors. (b) Image-based ranking signals, which output scores without a retention scale and are therefore evaluated by ranking metrics. Bold indicates the best value in each column.

(a) Retention regressors

Method	MAE↓	RMSE↓	AUROC↑	AP↑	AURC↓
HGB	3.297	6.332	96.503	90.897	5.237
XGBoost	3.409	6.508	96.327	90.455	5.299
CatBoost	3.402	6.507	96.385	90.427	5.266
Ridge	4.149	6.898	95.992	90.013	5.428
TabM	**2.772**	5.913	97.266	93.263	**5.038**
DINOv2 + head	3.532	6.542	95.540	90.216	5.678
CheXzero + head	4.308	8.526	87.942	79.770	9.749
BiomedCLIP + head	4.783	8.913	89.869	78.253	8.052
Global DenseNet	2.982	6.167	96.929	92.686	5.201
Mask geometry + HGB	5.719	8.532	87.662	75.708	8.743
OA-CXR	2.810	**5.856**	**97.445**	**93.931**	5.055

(b) Image-based ranking signals

Method	AUROC↑	AP↑	AURC↓
CheXpert-only	51.024	29.753	27.489
AutoQC CXR-NIQE	55.972	31.052	23.791
BioViL-T	46.607	24.584	29.314
OA-CXR	**97.445**	**93.931**	**5.055**

OA-CXR allocates review effort almost optimally (Table 9). At budgets of 5–30%, its recall equals the oracle bound, meaning that every report it selects contains a low-retention event. At a 50% budget, it captures 91.7% of event-positive reports (95% CI: 83.6–96.9%), or 96.9% of the oracle bound, whereas mean token log probability captures 50.0%, a paired gain of 41.7 percentage points (95% CI: 33.1–49.1). Mean token log probability tracks the random-allocation line at every budget, and it leaves the unreviewed reports with the base event rate (52.8%), whereas OA-CXR reduces this proportion to 8.8%. Equivalently, under random allocation, clinicians would need to review about 92% of eligible reports to reach, in expectation, the recall that OA-CXR attains by reviewing half of them.

Two further comparisons support the design. Mapping each statement to its finding-specific region, rather than scoring all statements with whole-lung retention, improves recall at the three larger budgets, by 2.6 percentage points at 50% (95% CI: 0.2–3.7), confirming the value of the knowledge base. Adding generation confidence through the calibrated combination reproduces the finding-specific recall exactly at all five budgets, so this row is omitted from Table 9; confidence carries no additional information about evidence sufficiency.

5.4. RQ4: External Validation on an Independent Clinical Cohort

The external cohort tests transfer to a different patient population, hospital, and annotation process: 31,083 queries from 797 pediatric radiographs with clinician-annotated lung masks, of which 27.63% are low-retention events. All models were frozen. OA-CXR achieves MAE 0.0281 (95% CI: 0.0272–0.0290), RMSE 0.0586, AUROC 0.974 (95% CI: 0.972–0.976), AP 0.939, and AURC 0.0506 (Table 10, Fig. 6). These errors are comparable to those on the public datasets, indicating that the representation transfers without adaptation. Relative to the global DenseNet control, OA-CXR reduces MAE by 5.77% and increases AUROC from 0.969 to 0.974, with a paired MAE interval favoring OA-CXR; the benefit of anatomical modeling thus persists under external shift.

The two readouts of the OA-CXR representation, the residual readout and TabM, are the strongest configurations, and both clearly outperform the frozen foundation-model encoders. Their overall MAE is similar (0.0281 versus 0.0277), with a paired difference interval that includes zero. OA-CXR achieves higher AUROC (0.974 versus 0.973) and AP (0.939 versus 0.933), with paired intervals favoring OA-CXR, and the AUROC difference remains significant after Holm adjustment (Fig. 6f; $P < 0.05$). Because budget-constrained review depends on ranking, this advantage bears directly on report prioritization. Regionally, TabM has lower basal MAE, OA-CXR has lower inner-rim MAE, and whole-lung MAE does not differ significantly after adjustment (Fig. 6g–l). OA-CXR also outperforms CheXpert-only, AutoQC CXR-NIQE, and BioViL-T on AUROC, AP, and AURC (Fig. 6a–c). Figure 6(d) shows six basal-retention examples on either side of the 0.9 threshold, with absolute errors of 0.0010–0.0089, together with input-gradient maps of the readout that indicate which image regions influence each estimate; in several cases, sensitivity is highest near the predicted basal supports.

5.5. Stress Test Beyond the Training Protocol

To probe behavior outside the training protocol, we constructed a stress test with 21 conditions per source image: the original; single-edge crops of 5% and 30%, which lie outside the training range; asymmetric opposing-edge crops of 7% and 23%; background-only cropping; brightness scaling by 0.7 and 1.3; Gaussian blur (radius 2); and padding modifications. Across the 8,805 public evaluation images, 184,864 valid presentations yield 554,592 queries; 41 padding conditions are inapplicable because the required padding or placement change cannot be made. Transformations that preserve the retained reference support leave the target unchanged.

Error increases for all methods under these unseen conditions, but OA-CXR remains the most accurate. Its stress-test macro MAE is 0.0771 (standard set: 0.0296), compared with 0.0874 for TabM (0.0308), 0.0875 for the global DenseNet control (0.0323), and 0.1162 for the mask-geometry control (0.0622); the OA-CXR error is thus 11.8% and 11.9% lower than those of TabM and the global control, respectively. Because TabM uses the same frozen features, this advantage reflects the robustness of the residual readout, whereas the margin over the global control reflects that of anatomical modeling.

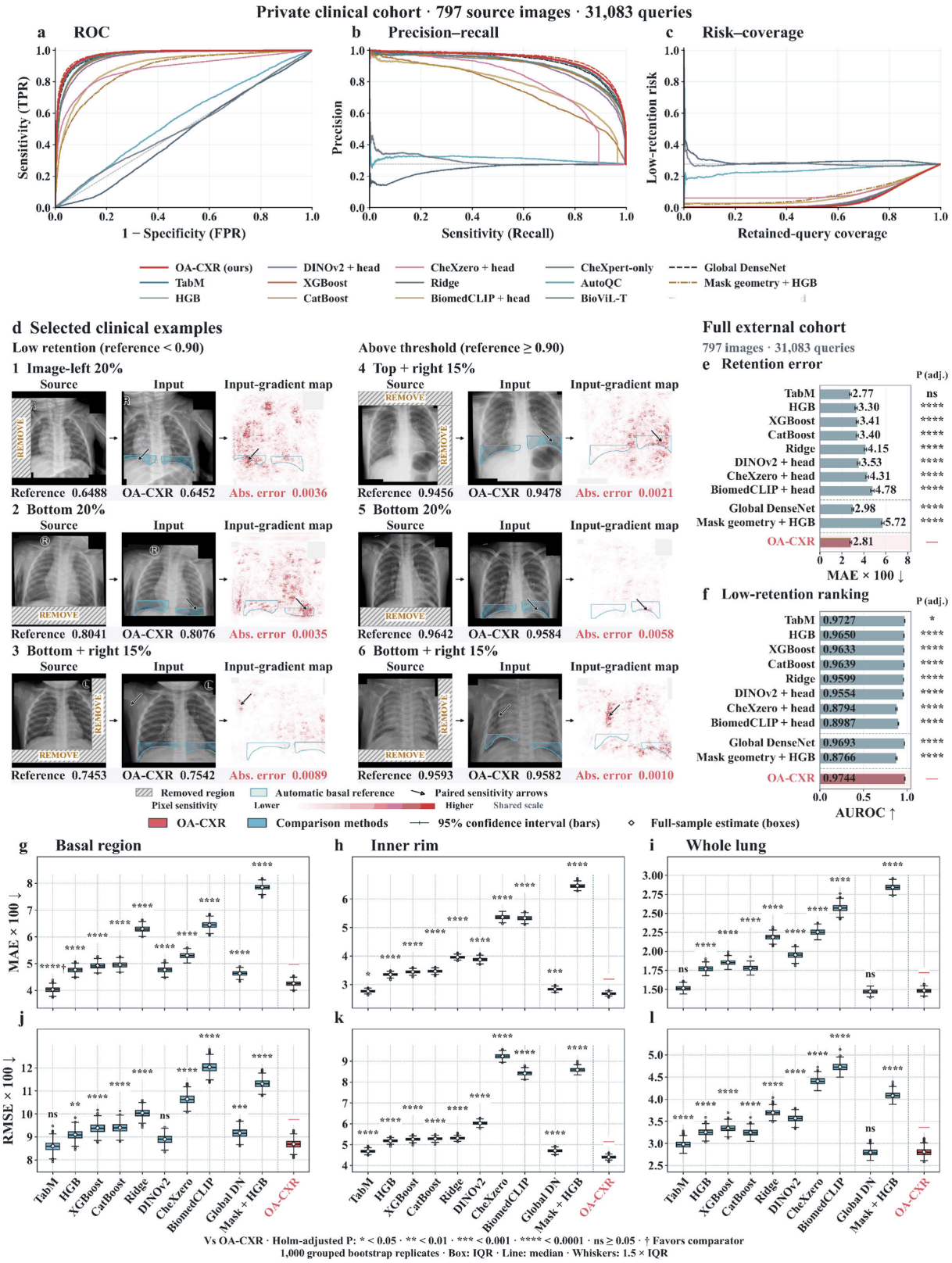


Figure 6: External validation on 797 pediatric radiographs (31,083 queries) against clinician-annotated references. **(a–c)** Low-retention detection (reference < 0.90). **(d)** Six illustrative cases with input-gradient maps. **(e,f)** Overall MAE and AUROC. **(g–l)** Regional MAE and RMSE (×100). Error bars show 95% CIs; boxplots summarize 1,000 paired bootstrap resamples of 791 identity groups. Significance uses paired jackknife t-tests with Holm adjustment across 80 comparisons: $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$, $^{****}P < 0.0001$; ns, $P \geq 0.05$. † favors the comparator; other significant differences favor OA-CXR.

6. Discussion

6.1. Why Generation Confidence Misses Evidence Insufficiency

The most consistent finding of this study is that generation confidence is uninformative about anatomical evidence. Token log probability, sentence log probability, entropy, and semantic consistency all perform close to or below chance at the statement level (Table 7), confidence-based allocation is indistinguishable from random allocation at the report level (Table 9), and adding confidence to retention leaves report-level recall unchanged at every budget. We attribute this blind spot to the nature of the statements involved. Negative statements such as "no pleural effusion or pneumothorax" are among the most frequent and formulaic phrases in chest radiograph reports, so a language model can produce them with high likelihood whether or not the supporting anatomy is visible. Confidence measures the plausibility of the text; evidence sufficiency measures the content of the image. The two are complementary, and a deferral policy that relies only on confidence leaves this class of unsupported statements unreviewed.

This observation has implications beyond OA-CXR. Learning-to-defer methods and selective predictors usually derive deferral scores from the predictive distribution of the model being audited (Geifman & El-Yaniv, 2017; Hendrickx et al., 2024; Mozannar & Sontag, 2020). Our results show that, for generative reporting, some clinically important failures leave the predictive distribution unchanged and can be detected only by an external, task-specific signal that examines the input. This finding is consistent with the observation that human–AI combinations do not automatically outperform their components (Vaccaro et al., 2024): the benefit of deferral depends on whether the deferral signal targets the errors that matter.

6.2. Design Insights and Practical Implications

Three design lessons emerge from RQ1 and RQ2. First, task adaptation matters more than encoder scale: the fine-tuned DenseNet-121 representation outperforms the frozen BiomedCLIP, CheXzero, and DINOv2 encoders on both the public and clinical cohorts, and freezing the encoder is the most damaging ablation. Second, global context, segmentation-grounded regional pooling, and learned features are complementary, whereas explicit geometry alone is insufficient. Third, explicit knowledge improves decisions without additional learning: finding-specific regions raise report-level recall over a whole-lung score and make every flag traceable to an inspectable rule.

In practice, OA-CXR acts as a post-generation quality gate between drafting and sign-off. It can run in a threshold mode, in which every flagged statement is reviewed (sensitivity 92.3% and specificity 95.8% at $\tau_{\text{review}} = 0.9$, adjustable to local risk tolerance), or in a budget mode, in which a fixed fraction of reports is reviewed in priority order. It needs neither the generator's internal states nor repeated sampling, so it can audit closed-source generators and survive generator updates. Its cost is small relative to the generators it audits: a single 256×256 encoder–decoder pass serves all three queries, followed by a two-block readout and rule-based parsing. New findings can be supported by adding a mapping to the knowledge base and, if needed, a region definition, such as apical regions for apical lesions.

6.3. Limitations and Future Work

First, retention is *source-relative*: it quantifies anatomy lost between the acquired radiograph and the presented image, not the completeness of the acquisition itself, and estimating absolute coverage would require priors on the expected extent of each region. Second, training and standard evaluation use controlled edge and corner crops; the stress test shows that unseen transformations increase error substantially, and training with broader presentation families is a natural next step. Third, training-source composition matters: in a supplementary analysis, a model trained without the pediatric Kermany/V7 images approximately doubled its MAE on Kermany/V7 evaluation images (0.0744 versus 0.0368), which motivates training on demographically diverse sources before deployment in new populations.

Fourth, report review covers explicit negative statements about three findings from one generator, and only 445 of 6,656 MAIRA-2 reports (6.7%) contained eligible statements; extending the knowledge base and replacing the rule-based parser with a validated clinical language model would broaden coverage. Fifth, the external cohort is pediatric and from a single institution, and a prospective reader study is needed to quantify clinical benefit. Finally, all models were fitted with a single seed, and duplicate screening of public sources relied on image fingerprints because patient-level identifiers were unavailable.

7. Conclusion

Generated chest radiograph reports can contain negative statements that are fluent but unsupported, because the anatomy on which they depend was not visible to the generator, and generation confidence cannot detect this failure. OA-CXR addresses it with source-relative anatomical retention: a knowledge base links each statement to its diagnostic region, and a two-stage global–regional network estimates how much of that region the presented image retains. Across four public datasets, OA-CXR estimated retention more accurately than nine alternative regressors and foundation-model encoders. In report review, it matched the oracle at budgets of up to 30% and captured 91.7% of event-positive reports by reviewing half of them, whereas confidence-based allocation performed at chance. On an independent pediatric cohort, the frozen system achieved MAE 0.0281 and AUROC 0.974. Anatomical evidence sufficiency is thus a practical, interpretable, and generator-independent signal for allocating clinician review, complementary to existing measures of report correctness.

Code and data availability

The source code and model-access instructions are available through an anonymous repository at *https://anonymous.4open.science/r/OA-CXR/*. The four public datasets are available from their respective sources, and download links and access instructions are provided in the repository. The independent pediatric clinical cohort from *[institution withheld for double-anonymized review]* is not

publicly available because of patient-privacy, confidentiality, and institutional data-protection requirements.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work, the authors used Claude (Anthropic) to improve the language and structure of the manuscript. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

References

Bannur, S., et al. (2023). Learning to exploit temporal structure for biomedical vision-language processing. *Proc. IEEE/CVF CVPR*, 15016–15027. https : / / openaccess . thecvf . com / content / CVPR2023/html/Bannur_Learning_To_Exploit_Temporal_Structure_for_Biomedical_Vision - Language_Processing_CVPR_2023_paper.html

Bannur, S., et al. (2024). MAIRA-2: Grounded radiology report generation. https://doi.org/10.48550/arXiv.2406.04449

Chen, T., & Guestrin, C. (2016). XGBoost: A scalable tree boosting system. *Proc. ACM SIGKDD*, 785–794. https://doi.org/10.1145/2939672.2939785

Chen, Z., et al. (2024). CheXagent: Towards a foundation model for chest X-Ray interpretation. https://doi.org/10.48550/arXiv.2401.12208

Cohen, J. P., et al. (2022). TorchXRayVision: A library of chest X-ray datasets and models. *Proc. Medical Imaging with Deep Learning*, *172*, 231–249. https://proceedings.mlr.press/v172/cohen22a.html

Dvijotham, K., Winkens, J., Barsbey, M., Ghaisas, S., Stanforth, R., Pawlowski, N., Strachan, P., Ahmed, Z., Azizi, S., Bachrach, Y., Culp, L., Daswani, M., Freyberg, J., Kelly, C., Kiraly, A., Kohlberger, T., McKinney, S., Mustafa, B., Natarajan, V., … Karthikesalingam, A. (2023). Enhancing the reliability and accuracy of AI-enabled diagnosis via complementarity-driven deferral to clinicians. *Nature Medicine*, *29*(7), 1814–1820. https://doi.org/10.1038/s41591-023-02437-x

European Commission. (1996). *European guidelines on quality criteria for diagnostic radiographic images*. Office for Official Publications of the European Communities.

Gaggion, N., Mosquera, C., Aineseder, M., Mansilla, L., Milone, D., & Ferrante, E. (2025). CheXmask Database: A large-scale dataset of anatomical segmentation masks for chest X-ray images [Version 1.0.0]. *PhysioNet*. https://doi.org/10.13026/3705-zg36

Gaggion, N., Mansilla, L., Mosquera, C., Milone, D. H., & Ferrante, E. (2023). Improving anatomical plausibility in medical image segmentation via hybrid graph neural networks: Applications to chest X-Ray analysis. *IEEE Transactions on Medical Imaging*, *42*(2), 546–556. https://doi.org/10.1109/TMI.2022.3224660

Gaggion, N., Mosquera, C., Mansilla, L., Saidman, J. M., Aineseder, M., Milone, D. H., & Ferrante, E. (2024). CheXmask: A large-scale dataset of anatomical segmentation masks for multi-center chest X-ray images. *Scientific Data*, *11*, 511. https://doi.org/10.1038/s41597-024-03358-1

Geifman, Y., & El-Yaniv, R. (2017). Selective classification for deep neural networks. *Advances in Neural Information Processing Systems*, *30*. https : / / proceedings . neurips . cc / paper / 2017 / hash / 4a8423d5e91fda00bb7e46540e2b0cf1 - Abstract.html

Gorishniy, Y., Kotelnikov, A., & Babenko, A. (2025). TabM: Advancing tabular deep learning with parameter-efficient ensembling. *Proc. ICLR*. https://proceedings.iclr.cc/paper_files/paper/2025/hash/c1ba41c694834aeef91ae161711d4939 - Abstract - Conference . html

Han, P., Li, X., Jing, S., & Wei, J. (2026). Dynamic feature fusion guiding and multimodal large language model refining for medical image report generation. *Expert Systems with Applications*, *299*, Article 130082. https://doi.org/10.1016/j.eswa.2025.130082

Hardy, R., Kim, S. E., Ro, D. H., & Rajpurkar, P. (2025). ReXTrust: A model for fine-grained hallucination detection in AI-generated radiology reports. *Proceedings of The First AAAI Bridge Program on AI for Medicine and Healthcare*, *281*, 173–182. https://proceedings.mlr.press/v281/hardy25a.html

Heiman, A., Zhang, X., Chen, E., Kim, S. E., & Rajpurkar, P. (2025). FactCheXcker: Mitigating measurement hallucinations in chest X-ray report generation models. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 30787–30796.

Hendrickx, K., Perini, L., Van der Plas, D., Meert, W., & Davis, J. (2024). Machine learning with a reject option: A survey. *Machine Learning*, *113*, 3073–3110. https://doi.org/10.1007/s10994-024-06534-x

Huang, G., Liu, Z., van der Maaten, L., & Weinberger, K. Q. (2017). Densely connected convolutional networks. *Proc. IEEE CVPR*, 4700–4708. https://openaccess.thecvf.com/content_cvpr_2017/html / Huang _ Densely _ Connected _ Convolutional _ CVPR _ 2017_paper.html

Huang, L., Islam, T., Miron, A., Hone, K., & Li, Y. (2026). Automated radiology report generation: A comprehensive review. *Expert Systems with Applications*, *331*, Article 133320. https://doi.org/10.1016/j.eswa.2026.133320

Irvin, J., et al. (2019). CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison. *Proc. AAAI*, *33*, 590–597. https://doi.org/10.1609/aaai.v33i01.3301590

Jaeger, S., et al. (2014). Two public chest X-ray datasets for computer-aided screening of pulmonary diseases. *Quantitative Imaging in Medicine and Surgery*, *4*(6), 475–477. https://doi.org/10.3978/j.issn.2223-4292.2014.11.20

Jain, S., et al. (2021). RadGraph: Extracting clinical entities and relations from radiology reports. *Proc. NeurIPS Datasets and Benchmarks*, *1*. https://datasets-benchmarks-proceedings.neurips.cc/paper/2021 / hash / c8ffe9a587b126f152ed3d89a146b445 - Abstract - round1.html

Kermany, D., Zhang, K., & Goldbaum, M. (2018). Labeled optical coherence tomography (OCT) and chest X-Ray images for classification. https://doi.org/10.17632/rscbjbr9sj.2

Leibig, C., Brehmer, M., Bunk, S., Byng, D., Pinker, K., & Umutlu, L. (2022). Combining the strengths of radiologists and AI for breast cancer screening: A retrospective analysis. *The Lancet Digital Health*, *4*(7), e507–e519. https://doi.org/10.1016/S2589-7500(22)00070-X

Liu, K., Ma, Z., Liu, M., Jiao, Z., Kang, X., Miao, Q., & Xie, K. (2026). Factual serialization enhancement: A key innovation for chest X-ray report generation. *Expert Systems with Applications*, *324*, Article 132550. https://doi.org/10.1016/j.eswa.2026.132550

Liu, X., Liu, W., & Zeng, R. (2026). ThoraxAgent: An intelligent agent-based decision support system for chest radiography analysis with cross-modal dynamic retrieval. *Expert Systems with Applications*, *322*, Article 132243. https://doi.org/10.1016/j.eswa.2026.132243

Meng, Y., Ruan, J., Yang, B., Gao, Y., Jin, J., Dong, F., Ji, H., He, L., Cheng, G., & Gong, X. (2022). Automated quality assessment of chest radiographs based on deep learning and linear regression cascade algorithms. *European Radiology*, *32*(11), 7680–7690. https://doi.org/10.1007/s00330-022-08771-x

Mittal, A., Soundararajan, R., & Bovik, A. C. (2013). Making a "completely blind" image quality analyzer. *IEEE Signal Processing Letters*, *20*(3), 209–212. https://doi.org/10.1109/LSP.2012.2227726

Mozannar, H., & Sontag, D. (2020). Consistent estimators for learning to defer to an expert. *Proceedings of the 37th International Conference on Machine Learning*, *119*, 7076–7087. https : // proceedings.mlr.press/v119/mozannar20b.html

Oquab, M., et al. (2024). DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*. https://openreview.net/forum?id=a68SUt6zFt

Ostmeier, S., Xu, J., Chen, Z., Varma, M., Blankemeier, L., Bluethgen, C., Michalson, A. E., Moseley, M., Langlotz, C., Chaudhari, A. S., & Delbrouck, J.-B. (2024). GREEN: Generative radiology report evaluation and error notation. *Findings of the Association for Computational Linguistics: EMNLP 2024*, 374–390. https://doi.org/10.18653/v1/2024.findings-emnlp.21

Pedregosa, F., et al. (2011). Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, *12*(85), 2825–2830. https://jmlr.org/papers/v12/pedregosa11a.html

Pollard, T., Moody, B. E., Lehman, L.-w. H., Gow, B. J., Fernandes, C., Xie, C., Johnson, A., Mark, R. G., & Heldt, T. (2026). PhysioNet as a global platform for biomedical research. *Nature Health*, *1*(8), 792–795. https://doi.org/10.1038/s44360-026-00096-z

Prokhorenkova, L., Gusev, G., Vorobev, A., Dorogush, A. V., & Gulin, A. (2018). CatBoost: Unbiased boosting with categorical features. *Advances in Neural Information Processing Systems*, *31*. https : / / proceedings . neurips . cc / paper / 2018 / hash / 14491b756b3a51daac41c24863285549-Abstract.html

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., Krueger, G., & Sutskever, I. (2021). Learning transferable visual models from natural language supervision. *Proceedings of the 38th International Conference on Machine Learning*, *139*, 8748–8763.

Selby, I. A., et al. (2025). A pipeline for automated quality control of chest radiographs. *Radiology: Artificial Intelligence*, *7*(3), e240003. https://doi.org/10.1148/ryai.240003

Sellergren, A., Kazemzadeh, S., Jaroensri, T., Kiraly, A., Traverse, M., Yang, L., et al. (2025). MedGemma technical report. https://doi.org/10.48550/arXiv.2507.05201

Stirenko, S., et al. (2018). Chest X-Ray analysis of tuberculosis by deep learning with segmentation and augmentation. *Proc. IEEE ELNANO*, 422–428. https://doi.org/10.1109/ELNANO.2018.8477564

Tahir, A. M., et al. (2021a). COVID-19 infection localization and severity grading from chest X-ray images. *Computers in Biology and Medicine*, *139*, 105002. https://doi.org/10.1016/j.compbiomed.2021.105002

Tahir, A. M., Chowdhury, M. E. H., Qiblawey, Y., Khandakar, A., Rahman, T., Kiranyaz, S., Khurshid, U., Ibtehaz, N., Mahmud, S., & Ezeddin, M. (2021b). COVID-QU-Ex. https://doi.org/10.34740/kaggle/dsv/3122958

Tanida, T., Müller, P., Kaissis, G., & Rueckert, D. (2023). Interactive and explainable region-guided radiology report generation. *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 7433–7442. https : / / doi . org / 10 . 1109 / CVPR52729.2023.00718

Tanno, R., Barrett, D. G. T., Sellergren, A., Ghaisas, S., Dathathri, S., See, A., Welbl, J., Lau, C., Tu, T., Azizi, S., Singhal, K., Schaekermann, M., May, R., Lee, R., Man, S., Mahdavi, S., Ahmed, Z., Matias, Y., Barral, J., … Ktena, I. (2025). Collaboration between clinicians and vision–language models in radiology report generation. *Nature Medicine*, *31*(2), 599–608. https://doi.org/10.1038/s41591-024-03302-1

Tiu, E., Talius, E., Patel, P., Langlotz, C. P., Ng, A. Y., & Rajpurkar, P. (2022). Expert-level detection of pathologies from unannotated chest X-ray images via self-supervised learning. *Nature Biomedical Engineering*, *6*(12), 1399–1406. https://doi.org/10.1038/s41551-022-00936-9

V7 Labs. (n.d.). COVID-19 X-ray dataset [Accessed: Sep. 25, 2026; publication year not specified]. https://github.com/v7labs/covid-19-xray-dataset

Vaccaro, M., Almaatouq, A., & Malone, T. (2024). When combinations of humans and AI are useful: A systematic review and meta-analysis. *Nature Human Behaviour*, *8*(12), 2293–2303. https://doi.org/10.1038/s41562-024-02024-1

Wang, C., Zhou, W., Ghosh, S., Batmanghelich, K., & Li, W. (2025). Semantic consistency-based uncertainty quantification for factuality in radiology report generation. *Findings of the Association for Computational Linguistics: NAACL 2025*, 1739–1754. https://doi.org/10.18653/v1/2025.findings-naacl.95

Wang, X., Peng, Y., Lu, L., Lu, Z., Bagheri, M., & Summers, R. M. (2017). ChestX-ray8: Hospital-scale chest X-Ray database and benchmarks on weakly-supervised classification and localization of common thorax diseases. *Proc. IEEE CVPR*, 2097–2106. https://openaccess.thecvf.com/content_cvpr_2017/html/Wang_ChestX-ray8_Hospital-Scale_Chest_CVPR_2017_paper.html

Zambrano Chaves, J. M., Huang, S.-C., Xu, Y., Xu, H., Usuyama, N., Zhang, S., Poon, H., et al. (2025). A clinically accessible small multimodal radiology model and evaluation metric for chest X-ray findings. *Nature Communications*, *16*, Article 3108. https://doi.org/10.1038/s41467-025-58344-x

Zeiser, F. A., da Costa, C. A., de Oliveira Ramos, G., Maier, A., & da Rosa Righi, R. (2024). CheXReport: A transformer-based architecture to generate chest X-ray reports suggestions. *Expert Systems with Applications*, *255*, Article 124644. https://doi.org/10.1016/j.eswa.2024.124644

Zhang, S., Sambara, S., Banerjee, O., Acosta, J. N., Fahrner, L. J., & Rajpurkar, P. (2025). RadFlag: A black-box hallucination detection method for medical vision language models. *Proceedings of the 4th Machine Learning for Health Symposium*, *259*, 1087–1103. https://proceedings.mlr.press/v259/zhang25c.html

Zhang, S., et al. (2023). BiomedCLIP: A multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs. https://doi.org/10.48550/arXiv.2303.00915

Zhu, Z., Zhang, Y., Zhuang, X., Zhang, F., Wan, Z., Chen, Y., Long, Q., Zheng, Y., & Wu, X. (2025). Can we trust AI doctors? A survey of medical hallucination in large language and large vision-language models. *Findings of the Association for Computational Linguistics: ACL 2025*, 6748–6769. https://doi.org/10.18653/v1/2025.findings-acl.350